Soundness of tableau derivability for entailment

The Open Logic Project contributors

openlogic:first-order-logic/tableaux:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author corollary labelled cor:entailment-soundness (2504-2507); the proof span 2509-2521 is the author proof environment that immediately follows it.

Editorial note. The source's iftag FOL/PL inline variants are retained as inline alternatives in the packaging; the FOL branch is the counted reading for this unit.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/tableaux, counted once: soundness of tableau derivability for entailment. Only the counted corollary and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Tableau derivability from a set. Gamma proves A in the tableau calculus when some finite {B_1,...,B_n} subset of Gamma has a closed tableau for {False A, True B_1,...,True B_n}.. It is the hypothesis the proof unpacks (2509-2513).
- Tableau soundness. The unit's soundness theorem: a closed tableau for a finite set of signed formulas cannot be satisfied, i.e., no structure satisfies all its members.. It converts the closed tableau into the satisfaction disjunction (2513-2518).

Source reference. Original theorem thm-tableau-soundness: sources/olp-first-order-logic-tableaux/pinned-original-source.tex, source lines 2359–2359. The cited result is separately proved within the same frozen unit and delivered as its own corpus card; the reference is contextual packaging, and the counted proof retains its own argumentative steps in full, so the delivery does not depend on the referenced statement for its completeness.

Corollary 0.1. *If* $\Gamma \vdash !A$ *then* $\Gamma \models !A$.

Proof. If $\Gamma \vdash !A$ then for some $!B_1$, ..., $!B_n \in \Gamma$, $\{\bot(!A), \top(!B_1), \ldots, \top(!B_n)\}$ has a closed !!tableau. By thm-tableau-soundness, every !!structure $\mathfrak{A}M$ either makes some $!B_i$ false or makes $!A$ true. Hence, if $M \models$ then also $M \models$. □

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